

Bill of Materials - VBF-T1R1-BOM-01

Bill of Materials (g/cell and kg/kWh; dual caliber)

Component	Mass (g/cell)	kg/kWh	Source / formula
Positive active material (NMC811)	16.8422	0.84349	75.6 um x 0.1027 m2 x 0.665 x 3262 kg/m3 (parameter set + override)
Positive conductive additive (carbon black)	0.2961	0.01483	literature default 96.7/1.7/1.6 wt% AM/CB/binder (estimate, annotated)
Positive binder (PVDF)	0.2787	0.01396	literature default 96.7/1.7/1.6 wt% AM/CB/binder (estimate, annotated)
Negative active material (graphite)	12.2525	0.61363	120 um x 0.1027 m2 x 0.60 x 1657 kg/m3 (parameter set + override)
Negative conductive additive (carbon black)	0.1276	0.00639	literature default 96/1/3 wt% AM/CB/binder (estimate, annotated)
Negative binder (CMC/SBR)	0.3829	0.01918	literature default 96/1/3 wt% AM/CB/binder (estimate, annotated)
Separator	0.1297	0.00649	6 um x 0.1027 m2 x (1-0.47) x 397 kg/m3 (parameter set + override)
Electrolyte	9.3842	0.46998	pore volume 7.820 cm3 x 1.2 g/cm3, fill factor 1.0 (literature density, estimate)
Positive current collector (Al)	2.2183	0.11110	8 um x 0.1027 m2 x 2700 kg/m3
Negative current collector (Cu)	4.6010	0.23042	5 um x 0.1027 m2 x 8960 kg/m3
Enclosure and tabs	Not modeled	-	Not modeled (no parameter)
Total (excl. enclosure/tabs)	46.5131	2.32946	sum of modeled components; cell energy 0.0199673 kWh (cell/r3_I_energy.json)